DRAWING DESCRIPTIONS — PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture block diagram showing the eight-layer self-sustaining neural-motor energy harvesting loop.

FIG. 2 is a comparative energy balance chart showing control system energy depletion versus experimental system energy growth over 2000 operational steps.

FIG. 3 is a schematic diagram of the spiking neural network architecture showing the leaky integrate-and-fire neuron model with synaptic connections.

FIG. 4 is a circuit schematic of the energy harvesting subsystem comprising piezoelectric and thermoelectric transducers with rectification and storage.

FIG. 5 is a graph showing the activity-dependent energy dynamics curve illustrating the quadratic cost model and optimal activity operating region.

FIG. 6 is a hardware reference design layout diagram showing component placement and interconnections for the physical embodiment.

FIG. 7 is a validation results summary table confirming all falsifiable claims pass the specified thresholds.

FIG. 8 is a reference architecture diagram of the energy-bounded recursive control system showing the canonical signal flow from sensor interface through neuromorphic core, recursive reflection, energy governor, cognitive modulation, neural router, and actuation, with energy harvesting feedback.

DETAILED DESCRIPTION OF THE DRAWINGS

FIG. 1 — System Architecture Block Diagram (8-Layer Loop)

This figure shows eight labeled blocks arranged in a vertical flow with arrows indicating signal direction, forming a closed loop.

Block layout (top to bottom, then looping back):

ENVIRONMENT (external, at top)

|

v

[L1: Printed Membrane] — labeled "ThermochromicMixin"

|

v

[L2: Sensing Pads] — labeled "light_intensity, membrane_temp"

|

v

[L3: Optical Transmission] — labeled "signal_voltage = light / 1000 x 5.0"

|

v

[L4: Neuromorphic CPU (SNN)] — labeled "BaseSNN (LIF neurons)"

| Arrow entering from L8 labeled

| "reflection feedback"

v

[L5: Servo Actuation] — labeled "target_angle = clip(output x 180)"

|

+-----> [L6: Energy Harvesting] — labeled "piezo + thermal"

|

v

[L7: Ground Reference] — labeled "noise floor / baseline"

|

v

[L8: Recursive Reflection] — labeled "previous_output"

|

+-----> (arrow loops back to L4)

Three signal pathways should be distinguished by line style:

- Solid line: Main spine (L1 through L8 to L4)

- Dashed line: Thermal cross-link (L1 membrane_temp directly to L6)

- Dotted line: Reflection feedback (L8 to L4)

All blocks are rectangles with rounded corners. Environment is shown as a cloud shape. The loop-back arrow from L8 to L4 should be prominent and labeled "THE LOOP CLOSES HERE."

FIG. 2 — Energy Balance Comparison

This figure shows a dual-panel line graph or single graph with two curves.

X-axis: Operational Steps (0 to 2000)

Y-axis: Cumulative Energy (mWh)

Curve 1 (Control — solid line, labeled "EnergyConfig (Control)"):

- Starts at 50.0 mWh

- Decreases linearly

- Ends at approximately -4,697 mWh at step 2000

- Label: "base_consumption = 470 mW"

Curve 2 (Experimental — dashed line, labeled "BalancedEnergyConfig (Experimental)"):

- Starts at 50.0 mWh

- Rises rapidly to 100.0 mWh (storage capacity) within approximately 30 steps

- Plateaus at 100.0 mWh for remaining steps (homeostatic equilibrium)

- Label: "base_consumption = 45 mW, capacity = 100 mWh, overflow → heat"

A horizontal dashed line at y = 0 labeled "Self-Sustaining Threshold."

A horizontal dashed line at y = 100 labeled "Storage Capacity (100 mWh)."

Annotations:

- Arrow pointing to control curve: "System fails — energy depleted"

- Arrow pointing to experimental curve plateau: "System achieves homeostasis at capacity"

The area between the two curves should be lightly shaded to emphasize the divergence.

FIG. 3 — SNN Architecture (LIF Neuron Model)

This figure shows the internal structure of the spiking neural network.

Top section — Single LIF Neuron Detail:

- Circle labeled "Neuron i"

- Input arrows entering from left: "I_external" and "I_reflection"

- Inside the circle: equation written as "V(t+1) = V(t) x leak + I x dt"

- Threshold line: horizontal dashed line labeled "V_threshold = 0.5"

- Output arrow exiting right: "spike (0 or 1)"

- Below neuron: "If V >= threshold: spike, V = V_reset"

Bottom section — Network Topology:

- N circles (labeled "1" through "N") arranged in a single layer

- All-to-all connections shown as lines between neurons

- Weight matrix notation: "W[i,j]" on one representative connection

- Input arrow from left entering all neurons: "adjusted_input"

- Output arrows from all neurons going right: "spike_vector"

- Self-connections shown as small loops, crossed out (zeroed)

- Label: "N = 50 neurons (operational), N = 20 neurons (control)"

Inset box showing STDP rule:

- "If post fires: W += A+ x pre_trace (LTP)"

- "If pre fires: W -= A- x post_trace (LTD)"

- "A- > A+ (stability bias)"

FIG. 4 — Energy Harvesting Circuit

This figure shows an electrical circuit schematic.

Left side — Piezoelectric Harvester:

- Piezo disc symbol (capacitor with "PIEZO" label)

- Connected to servo shaft via mechanical coupling (zigzag line)

- Label: "27mm piezoelectric disc"

- Output leads to full-bridge rectifier (4 diodes in bridge configuration)

- Rectifier output to smoothing capacitor

- Label: "friction_factor = 18.0"

Center — LC Resonance Circuit:

- Inductor symbol in series with piezo output

- Label: "Laird ferrite-backed inductor"

- Resonance annotation: "f_res = 1 / (2 x pi x sqrt(L x C))"

- Label: "3-10x voltage amplification at resonance"

Right side — Thermoelectric Harvester:

- Thermistor/thermocouple symbol

- Label: "NTC 10K 3950"

- Connected to separate rectifier path

- Label: "thermal_factor = 8.0"

Bottom — Energy Storage:

- Both harvester outputs combine at a summing junction

- Arrow to storage element (battery/capacitor symbol)

- Label: "energy_stored (mWh)"

- Output arrow labeled "Power to SNN (L4)"

Power budget annotation box:

- "Harvest = friction_harvest + thermal_harvest"

- "Cost = base_consumption + activity_cost x spike_count^2"

- "Net = Harvest - Cost"

FIG. 5 — Activity-Dependent Energy Dynamics

This figure shows a single graph illustrating the energy sweet spot.

X-axis: "Neural Activity (spike_count / N)" ranging from 0.0 to 1.0

Y-axis: "Energy Rate (mWh/step)"

Three curves:

1. Harvest curve (solid, increasing): Linear function of activity

- Label: "E_harvest = friction_factor x angular_velocity + thermal_factor x delta_T"

- Starts near zero, increases linearly with activity

2. Cost curve (dashed, quadratic): Quadratic function of activity

- Label: "E_cost = base + activity_cost x (spikes/N)^2"

- Starts at base_consumption, curves upward quadratically

3. Net energy curve (bold): Harvest minus Cost

- Shows positive region (shaded green/light) at moderate activity

- Shows negative regions at very low and very high activity

- Peak labeled "Optimal Operating Point"

Vertical dashed lines marking regions:

- 0.0 to 0.2: "Low activity — insufficient harvesting"

- 0.2 to 0.7: "Sweet spot — net positive energy" (shaded region)

- 0.7 to 1.0: "High activity — quadratic cost dominates"

Annotation: "Self-sustaining operation requires maintaining activity in the sweet spot region"

FIG. 6 — Hardware Reference Design

This figure shows a top-down component layout diagram.

Central component:

- Rectangle labeled "Raspberry Pi Pico (RP2040)" with pin labels on edges

Connected components (arranged around the Pico):

Top left:

- Circle labeled "Piezo Disc 27mm"

- Connected to Pico via ADC pin (GP26/ADC0)

- Mechanical coupling line to Servo

Top right:

- Rectangle labeled "SG90 Micro Servo"

- Connected to Pico via PWM pin (GP0)

- Mechanical coupling to Piezo disc (zigzag line)

Bottom left:

- Small component labeled "NTC 10K Thermistor"

- Connected to Pico via ADC pin (GP27/ADC1)

Bottom center:

- Small component labeled "GL5528 LDR"

- Connected to Pico via ADC pin (GP28/ADC2)

Bottom right:

- Small square labeled "WS2812B RGB LED"

- Connected to Pico via digital pin (GP15)

Left side:

- Component labeled "Laird Ferrite Inductor"

- Connected in series between Piezo and rectifier

- Label: "LC resonance circuit"

Power section:

- Rectifier bridge symbol

- Storage capacitor

- Arrow labeled "Harvested energy feedback"

BOM annotation box in corner:

- "Pi Pico: $4"

- "Piezo 27mm: $2"

- "SG90 Servo: $3"

- "Thermistor: $0.15"

- "LDR: $0.10"

- "WS2812B: $0.25"

- "Laird Inductor: $0 (inventor stock)"

- "Total BOM: <$15"

FIG. 7 — Validation Results Summary

This figure shows a structured table summarizing all validated claims.

Table with columns: Claim | Criterion | Measured Value | Threshold | Result

Section header: "Phase 7 Control Baseline (EnergyConfig)"

Row A: Closed-loop continuity | All channels 2000 points | 2000 | 2000 | PASS

Row B: Boundedness | All state variables bounded | bounded | bounded | PASS

Row C: Robustness under noise | Std(theta) <= 45 deg | [value] | 45 | PASS

Row D: Input-output gain | corr(L,theta) >= 0.2 | [value] | 0.2 | PASS

Row E: Saturation ratio | sat_theta <= 0.20 | [value] | 0.20 | PASS

Section header: "Phase 7 Full Integration (Control confirms energy depletion)"

Row: Energy at 2000 steps | Energy depleted | -4,697 mWh | < 0 | CONFIRMED

Section header: "MCP Integration (BalancedEnergyConfig)"

Row: Energy at 2000 steps | Homeostasis at capacity | 100.0 mWh | = capacity | CONFIRMED

Row: Stability at 2,000,000 steps | Energy std in last 50% | 0.0000 mWh | < 1.0 | CONFIRMED

Section header: "Phase 8 STDP"

Row F8.1: Weight entropy decrease | final < initial | 2.95 to 2.02 bits | decrease | PASS

Row F8.2: STDP MI > 1.2x frozen | ratio | 6.52x | > 1.20 | PASS

Row F8.3: Late weight delta < 10% early | ratio | 0.00% | < 10% | PASS

Footer: "All claims validated. System is reproducible via git tags."

FIG. 8 — Energy-Bounded Recursive Control Architecture

This figure is the canonical embodiment and reference signal flow for the entire system. It shows the complete architecture as a vertically stacked block diagram with bidirectional control paths and energy feedback. This is the primary reference figure for patent examination.

System operation is constrained by real-time energy availability derived from physical interaction with the environment, thereby ensuring bounded, non-abstract execution.

Layout — Vertically stacked functional blocks with lateral connections:

TOP — Physical Environment:

■■■
■■■■■■■■■■■■■■

■ PHYSICAL ENVIRONMENT ■

■■■
■■■■■■■■■■■■■■

Two downward arrows from Environment:

Left arrow: "(Light / Temp / Force)" — to Sensor Interface

Right arrow: "(Mechanical Load)" — to Energy Harvesting

TIER 1 — Dual Input (left and right, same level):

Left block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ SENSOR INTERFACE ■

■ (BH1750, DS18B20, ■

■ Piezo Input) ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Specific sensor ICs for light (BH1750), temperature (DS18B20),

and force (piezo input)

Arrow down to Neuromorphic Core

Right block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ ENERGY HARVESTING ■

■ (Piezo → Rectifier → ■

■ Supercap) ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Physical energy transduction chain: piezoelectric element through

full-bridge rectifier to supercapacitor storage

Arrow down to Neuromorphic Core

TIER 2 — Central Processing (full width):

■■■

■■■■■■■■■■■■■■

■ NEUROMORPHIC CORE (SNN) ■

■ ■

■ • Spiking Neurons ■

■ • Membrane Dynamics ■

■ • Optional STDP ■

■ • Activity-Dependent Energy Cost ■

■■■

■■■■■■■■■■■■■■

Two downward arrows from Neuromorphic Core:

Left arrow: to Recursive Reflection

Right arrow: to Energy Governor

Internal labels:

- "Spiking Neurons" — leaky integrate-and-fire neural population

- "Membrane Dynamics" — $V(t) = V(t-1) \times \text{leak} + I_{adjusted}$

- "Optional STDP" — spike-timing dependent plasticity (opt-in)

- "Activity-Dependent Energy Cost" — quadratic: $\text{base} + k \times (\text{spikes}/N)^2$

TIER 3 — Dual Processing (left and right, same level, BIDIRECTIONAL):

Left block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ RECURSIVE REFLECTION ■

■ ■

■ • Prior State Memory ■

■ • Output Feedback ■

■ • Temporal Context ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Self-observation mechanism — feeds aggregate output at timestep t

back as input at timestep t+1, scaled by reflection coefficient

Right block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ ENERGY GOVERNOR ■

■ ■

■ • Storage Level ■

■ • Cost Function ■

■ • Activity Damping ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Energy constraint enforcement — monitors storage level, applies activity-dependent cost function, damps activity when energy is low

CRITICAL: Bidirectional arrow between Recursive Reflection and Energy Governor (■■■■■■■■■■)

Label: "Energy state modulates reflection gain; reflection state influences energy cost through activity level"

Both blocks have arrows downward to Cognitive Modulation Layer

TIER 4 — Modulation (full width):

■■■
■■■■■■■■■■■■■■

■ COGNITIVE MODULATION LAYER ■

■ ■

■ • Attention Weighting ■

■ • Intention Bias ■

■ • Contextual Suppression / Amplification ■

■ • (Optional) External Advisory Input ■

■■■
■■■■■■■■■■■■■■

Two downward arrows:

Left arrow: to Neural Router

Right arrow: to Telemetry & Logging

Internal labels:

- "Attention Weighting" — prioritizes input channels

- "Intention Bias" — goal-directed modulation

- "Contextual Suppression / Amplification" — state-dependent gating

- "(Optional) External Advisory Input" — external control layer when connected

TIER 5 — Output (left and right, same level):

Left block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ NEURAL ROUTER ■

■ ■

■ • Priority Arbitration■

■ • Energy Gating ■

■ • Congestion Control ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Signal arbitration — routes neural output to actuators based on priority, gates signals when energy is insufficient, prevents output saturation through congestion control

Arrow down to Actuation & Response

Right block:

■■■■■■■■■■■■■■■■■■■■■■■■■■

■ TELEMETRY & LOGGING ■

■ ■

■ • Energy ■

■ • State ■

■ • Reflection Metrics ■

■■■■■■■■■■■■■■■■■■■■■■■■■■

Label: Observability output — records energy level, system state, and

reflection metrics for validation and monitoring

No downward arrow (terminal output)

TIER 6 — Actuation (full width):

■■■
■■■■■■■■■■■■■■

■ ACTUATION & RESPONSE ■

■ ■

■ • Servo Control ■

■ • Mechanical Interaction ■

■ • Environmental Perturbation ■

■■■
■■■■■■■■■■■■■■

Label: Physical output — servo control drives mechanical interaction,

which perturbs the environment (closing the loop back to Physical

Environment at top) and generates mechanical load (feeding Energy

Harvesting at Tier 1)

Two upward feedback arrows from Actuation:

1. Arrow back to PHYSICAL ENVIRONMENT (top): "Environmental Perturbation"

2. Arrow back to ENERGY HARVESTING (Tier 1 right): "Mechanical Load

generates piezoelectric energy"

FEEDBACK LOOPS (mark each distinctly):

Loop A — Energy Harvesting Loop:

Actuation → Mechanical Load → Energy Harvesting → Neuromorphic Core

Style: Bold solid arrows

Label: "ENERGY LOOP — Motor activity generates power for neural computation"

Loop B — Recursive Reflection Loop:

Neuromorphic Core → Recursive Reflection → Neuromorphic Core

Style: Dotted arrows

Label: "REFLECTION LOOP — Prior state observation at configurable gain"

Loop C — Environmental Perturbation Loop:

Actuation → Environment → Sensor Interface → Neuromorphic Core

Style: Dashed arrows

Label: "SENSORY LOOP — Actions change environment, environment changes input"

Loop D — Energy Governor Feedback:

Energy Harvesting → Energy Governor ↔ Recursive Reflection → Cognitive Modulation

Style: Thin solid arrows

Label: "REGULATION LOOP — Energy availability constrains all processing"

Key annotations at bottom of figure:

"This architecture is physically grounded: all computation is constrained by real-time energy availability derived from the system's own mechanical interaction with its environment."

"No external power source is required for sustained operation under the balanced energy configuration."

FIG. 8 illustrates one exemplary embodiment; alternative embodiments may omit, combine, or substitute individual modules while preserving the energy-bounded recursive control loop.

Sheet numbering: "8/8" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/8", "2/8", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures